Ex:-13:-[B.Q:-13]:- A vendor sells 50% fruits he had & throws 20% of remaining. Next day he sells 60% of rem & throws away the rest. What % of his apples throwed totally?

Ans:- Total = 100%.
- 50% → sold
50% → Rem
- 10% → 20% of rem = 20% of 50% = 10% [throwed] — ①
40% → Rem
- 24% → 60% of rem = 60% of 40% = 24% [sold]
16% → throwed — ②

So totally throwed fruits = ① + ② = 10% + 16% = 26%.

[B.Q:-14]
Ex:-14:- Alisha purchased 60/- worth, out of which 40 paisa paid tax. If the tax rate is 10%. Then, what is tax-free paid amount?

Ans:- 10% tax = 40 paisa
100% = total amount paid with tax = 440 paisa = 4.40/-
tax-free amount = 60 - Taxable amount
= 60 - 4.4 = 55.6/-

Ex:-15:-[B.Q:-16]:- A got 30% of total marks & failed by 10 Marks. However B got 40% of total marks & got 15 marks more than pass marks. Pass Mark is?

Pass Marks w.r.t A = 30% + 10 = P
" " " B = 40% - 15 = P
⇒ 30% + 10 = 40% - 15
10% = 25 Marks
⇒ Pass Marks = 30% + 10 = 75 + 10 = 85

Ex:-16:-[B.Q:-17]:- Distribution of 2000 sweets among students, such a way that each student should get 20% of no. of students. Then how many sweets does each student have?

Ans:- Total students = x, sweets to each student = 0.2x
total sweets = 2000 = x × 0.2x ⇒ x = 100
∴ Each student sweets = 0.2x = 20

Ex:-17:- [B·Q:- 19]:- In an election b/w two, 10% of votes cast was invalid. The winner gets 70% of valid votes & won by 5400 votes. The total no. of votes are?

Ans:- Lets assume total votes = 100%.

10% → Invalid votes

90% votes — 90% valid votes

70% of valid votes — 30% of valid votes

70% of 90% − 30% of 90%

⇒ 63% − 27% = Majority = 5400 votes

⇒ 36% ⟶ 5400

1% ⟶ $\frac{5400}{36}$ = 150 ⇒ **100% = 15000**

Ex:-18:- [B·Q:- 21]:- In a class of 52 students, 25% are rich & the others are poor. There are 20 female in class out of which 55% are poor. Find the no. of rich males?

Ans:- 52

25% = $\frac{1}{4}$ × 52 = 13 rich

75% or 100% − 25% = 52 − 13 = 39 Poor

20 Female

55% = 55% of 20 = $\frac{11}{20}$ × 20 = 11 are poor

20 − 11 = 9 rich females

∴ Rich males = Total rich − Rich females

= 13 − 9 = **4**

Ex:-19:- [B·Q:-22]:- If the radius of circle ↑ed 5%, then area ↑ed by?

$C.A = \pi r^2$, $C.A_{new} = \pi(1.05r)^2$

$= \pi r^2(1.05)^2 = 1.1025\pi r^2$

$\%\text{change} = \frac{F-I}{I} \times 100$

$= \frac{1.1025\pi r^2 - \pi r^2}{\pi r^2} \times 100 = \frac{\pi r^2[0.1025]}{\pi r^2} \times 100$

= 10.25% ↑ed

(31)

Ex:-20:- [B.Q:-23] [B.Q]:- The mean annual salary paid to all employees is Rs 5000/-. The mean salary paid to all male & female workers 5200/- & 4200. The % of female workers?

Total salary of all employees = (M+F)5000

" " " " " = 5200M + 4200F

$\Rightarrow (M+F)5000 = 5200M + 4200F \Rightarrow$

$\Rightarrow 5000M + 5000F = 5200M + 4200F \Rightarrow 200M = 800F$

$\Rightarrow \frac{M}{F} = \frac{800}{200} \Rightarrow M:F = 4:1 \Rightarrow$ Total = 5 parts

$\text{Female \%} = \frac{F}{T} \times 100 = \frac{1}{5} \times 100 = \boxed{20\%}$

Ex:-21:- [B.Q:-25]:- The value of commodity depreciated by 20% annually. If the value of commodity 3 years ago was 10,500/-. Find its present value?

Ans:- Lets assume 3 yrs ago as 'x' = 10,500

2 yrs ago = ↓ed by 20% to 3yrs ago = 0.8x

1yr ago = " " " 2 " " = 0.8 × 0.8x

present = " " " 1 " " = 0.8 × 0.8 × 0.8x

Present price = = 0.8 × 0.8 × 0.8x

$\Rightarrow 0.8 \times 0.8 \times 0.8x = (0.8)^3 \times 10,500 = 0.512 \times 10,500$

$= 512 \times 10.5 = 5120 + 256 = \underline{5376/-}$